Hotel Bookings Dataset

Data Cleaning & Preparation Report

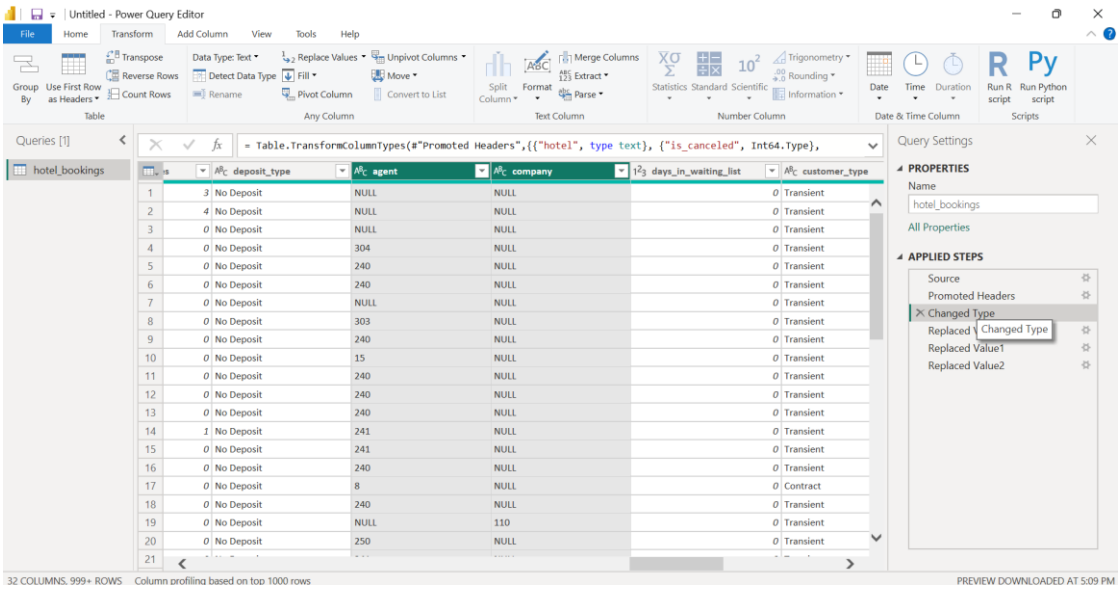
Tool Used:	Microsoft Power BI Desktop
Dataset:	Hotel Booking Demand (hotel_bookings)
Report Purpose:	Document all cleaning steps performed before dashboard building
Dashboard Pages:	Page 1: Executive Overview | Page 2: Cancellation Analysis | Page 3: Customer Insights

This report documents every data cleaning and transformation step applied to the hotel bookings dataset inside Power BI before building the 3-page analytics dashboard. Each section includes the steps taken, the DAX/Power Query logic used, and a placeholder for your screenshot.

Step 1: Handle NULL / Missing Values

Columns with NULL values identified:

agent — Many rows had NULL (meaning no agent was used for the booking) **company** — Many rows had NULL (meaning no company affiliation) **children** — Some rows had NULL instead of 0



Before cleaning

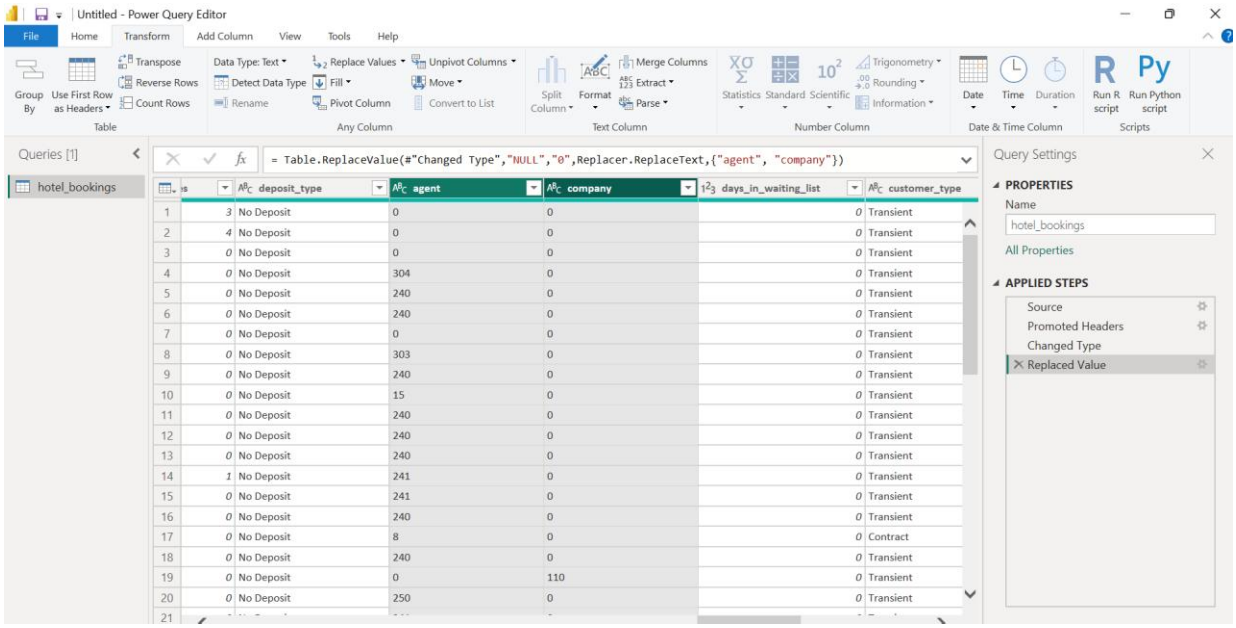
Transform Data → Select column → Replace Values → Replace null with 0

```
agent: null replaced with 0

company: null replaced with 0

children: null replaced with 0
```

Reason: NULL in agent/company means no agent or company was involved in the booking — replacing with 0 allows correct numeric aggregation.



[After cleaning]

Step 2: Fix Data Types

Correct data types were assigned to all columns in Power Query Editor (Transform Data → Data Type).

Column	Correct Data Type	Reason
is_canceled	Whole Number	Binary flag (0 or 1)
lead_time	Whole Number	Number of days
arrival_date_year	Whole Number	Year value
arrival_date_month	Text	Month name (January, etc.)
arrival_date_day_of_month	Whole Number	Day number
stays_in_weekend_nights	Whole Number	Count of nights
stays_in_week_nights	Whole Number	Count of nights

adults / children / babies	Whole Number	Guest counts
adr	Decimal Number	Average Daily Rate (price)
reservation_status_date	Date	Must be DATE not text
is_repeated_guest	Whole Number	Binary flag (0 or 1)

Critical: reservation_status_date must be set to Date type — errors may appear; choose 'Replace Current'.

Step 3: Remove Duplicate Rows

Duplicate rows were checked and removed using ALL columns — not based on a single column. A row is only a duplicate if every column value matches exactly.

Steps followed:

1. Opened Power Query Editor (Home → Transform Data)
2. Selected all columns using the top-left corner checkbox (Ctrl+A)
3. Clicked: Home → Remove Rows → Remove Duplicates
4. Checked row count before and after to confirm how many were removed

Important: Rows with lead_time = 72 appearing 3 times were inspected — they may be valid unique bookings. Only rows identical across ALL 32 columns were removed.

Step 4: Create New Calculated Columns (DAX)

The following new columns were created in Power BI using Modeling → New Column.

Total Nights

Purpose: Calculates how long each guest stayed

```
Total Nights = 'hotel_bookings'[stays_in_weekend_nights] +
'hotel_bookings'[stays_in_week_nights]
```

Total Guests

Purpose: Total number of people per booking

```
Total Guests = 'hotel_bookings'[adults] + 'hotel_bookings'[children] +
'hotel_bookings'[babies]
```

Cancellation Status

Purpose: Converts 0/1 to readable text for visuals

```
Cancellation Status = IF('hotel_bookings'[is_canceled] = 1, "Canceled", "Not
Canceled")
```

Booking Type

Purpose: Identifies new vs repeated guests

```
Booking Type = IF('hotel_bookings'[is_repeated_guest] = 1, "Repeated Guest", "New Guest")
```

Lead Time Category

Purpose: Groups bookings by how far in advance they were made

```
Lead Time Category = SWITCH(TRUE(), 'hotel_bookings'[lead_time] <= 30, "Short Lead", 'hotel_bookings'[lead_time] <= 90, "Medium Lead", "Long Lead")
```

Month Number

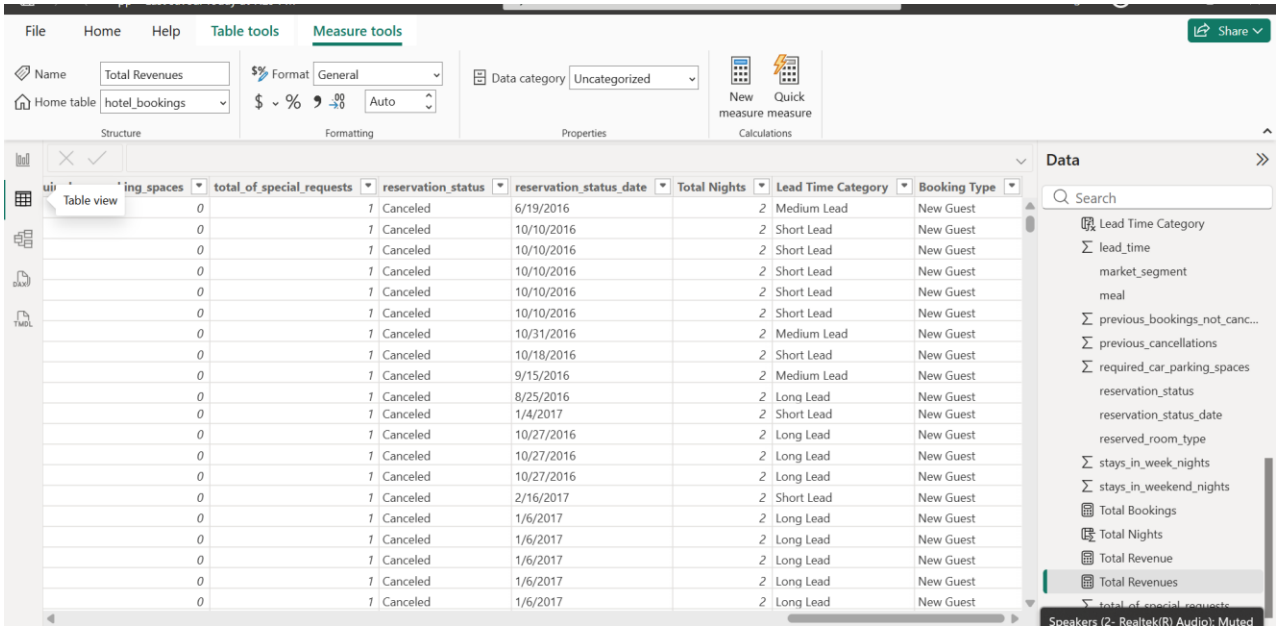
Purpose: Converts month name to number (1-12) for correct sorting

```
Month Number = SWITCH('hotel_bookings'[arrival_date_month], "January",1, "February",2, ... "December",12)
```

Arrival Date

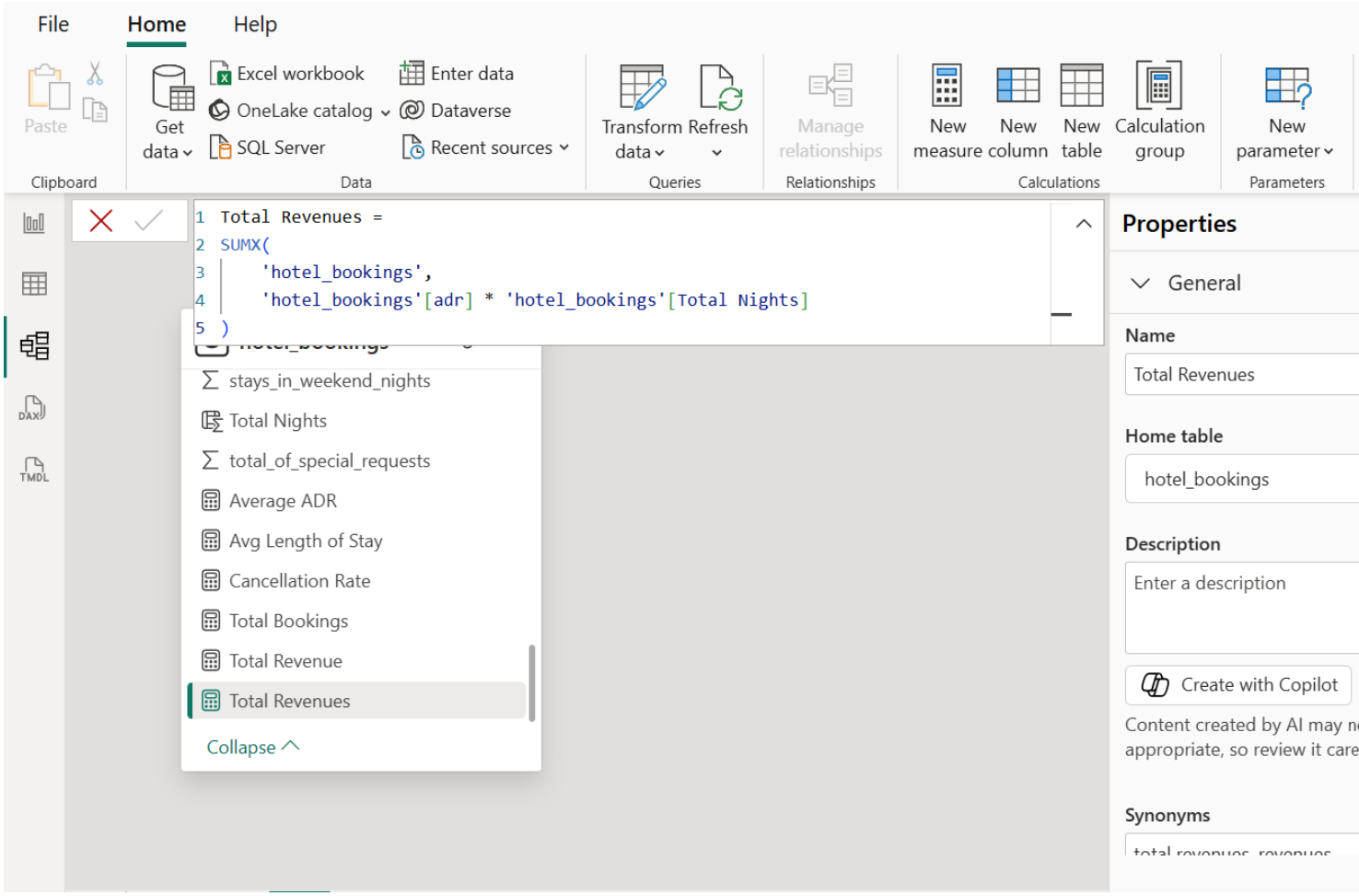
Purpose: Creates a proper date column from year + month + day

```
Arrival Date = DATE('hotel_bookings'[arrival_date_year], 'hotel_bookings'[Month Number], 'hotel_bookings'[arrival_date_day_of_month])
```



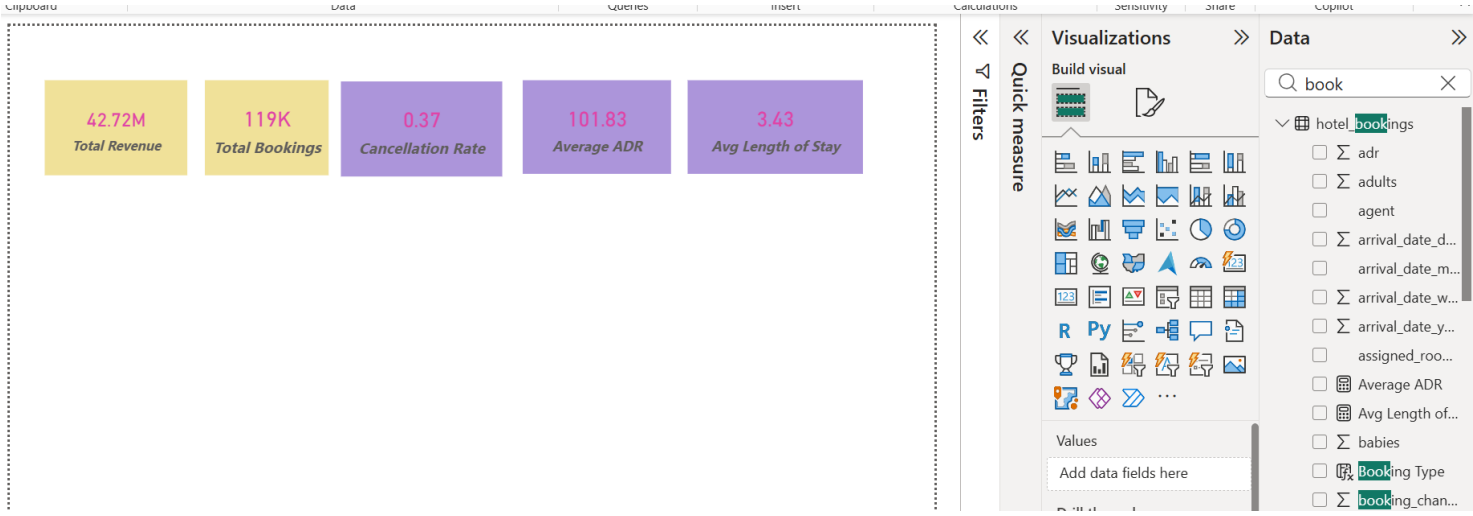
ing_spaces	total_of_special_requests	reservation_status	reservation_status_date	Total Nights	Lead Time Category	Booking Type
0	1	Canceled	6/19/2016	2	Medium Lead	New Guest
0	1	Canceled	10/10/2016	2	Short Lead	New Guest
0	1	Canceled	10/10/2016	2	Short Lead	New Guest
0	1	Canceled	10/10/2016	2	Short Lead	New Guest
0	1	Canceled	10/10/2016	2	Short Lead	New Guest
0	1	Canceled	10/10/2016	2	Short Lead	New Guest
0	1	Canceled	10/31/2016	2	Medium Lead	New Guest
0	1	Canceled	10/18/2016	2	Short Lead	New Guest
0	1	Canceled	9/15/2016	2	Medium Lead	New Guest
0	1	Canceled	8/25/2016	2	Long Lead	New Guest
0	1	Canceled	1/4/2017	2	Short Lead	New Guest
0	1	Canceled	10/27/2016	2	Long Lead	New Guest
0	1	Canceled	10/27/2016	2	Long Lead	New Guest
0	1	Canceled	10/27/2016	2	Long Lead	New Guest
0	1	Canceled	2/16/2017	2	Short Lead	New Guest
0	1	Canceled	1/6/2017	2	Long Lead	New Guest
0	1	Canceled	1/6/2017	2	Long Lead	New Guest
0	1	Canceled	1/6/2017	2	Long Lead	New Guest
0	1	Canceled	1/6/2017	2	Long Lead	New Guest
0	1	Canceled	1/6/2017	2	Long Lead	New Guest

Step 5: Create DAX Measures for Dashboard KPIs



Measures (not columns) were created for all KPI cards and chart calculations. Measures are more efficient and respond to filters dynamically.

Measure Name	DAX Formula
Total Revenue	`SUMX('hotel_bookings', 'hotel_bookings'[adr] * 'hotel_bookings'[Total`
Total Bookings	`COUNTROWS('hotel_bookings')`
Total Cancellations	`CALCULATE(COUNTROWS('hotel_bookings'), 'hotel_bookings'[is_canceled] = 1)`
Cancellation Rate	`DIVIDE([Total Cancellations], [Total Bookings], 0)` → Format as Percentage
Average ADR	`AVERAGE('hotel_bookings'[adr])` → Format as Currency
Avg Length of Stay	`AVERAGE('hotel_bookings'[Total Nights])` → Format as Decimal



Step 6: Clean & Standardize Text Columns

Text columns were cleaned to remove extra spaces and hidden characters using Power Query.

Columns cleaned:

- **hotel meal**
- **market_segment**
- **distribution_channel**
- **deposit_type**
- **customer_type**
- **reservation_status**

Steps in Power Query:

Select column → Transform tab → Format → Trim (removes leading/trailing spaces)

Select column → Transform tab → Format → Clean (removes hidden characters)

Step 7: Remove Invalid / Outlier Rows

Rows that do not represent valid bookings were filtered out to ensure data quality.

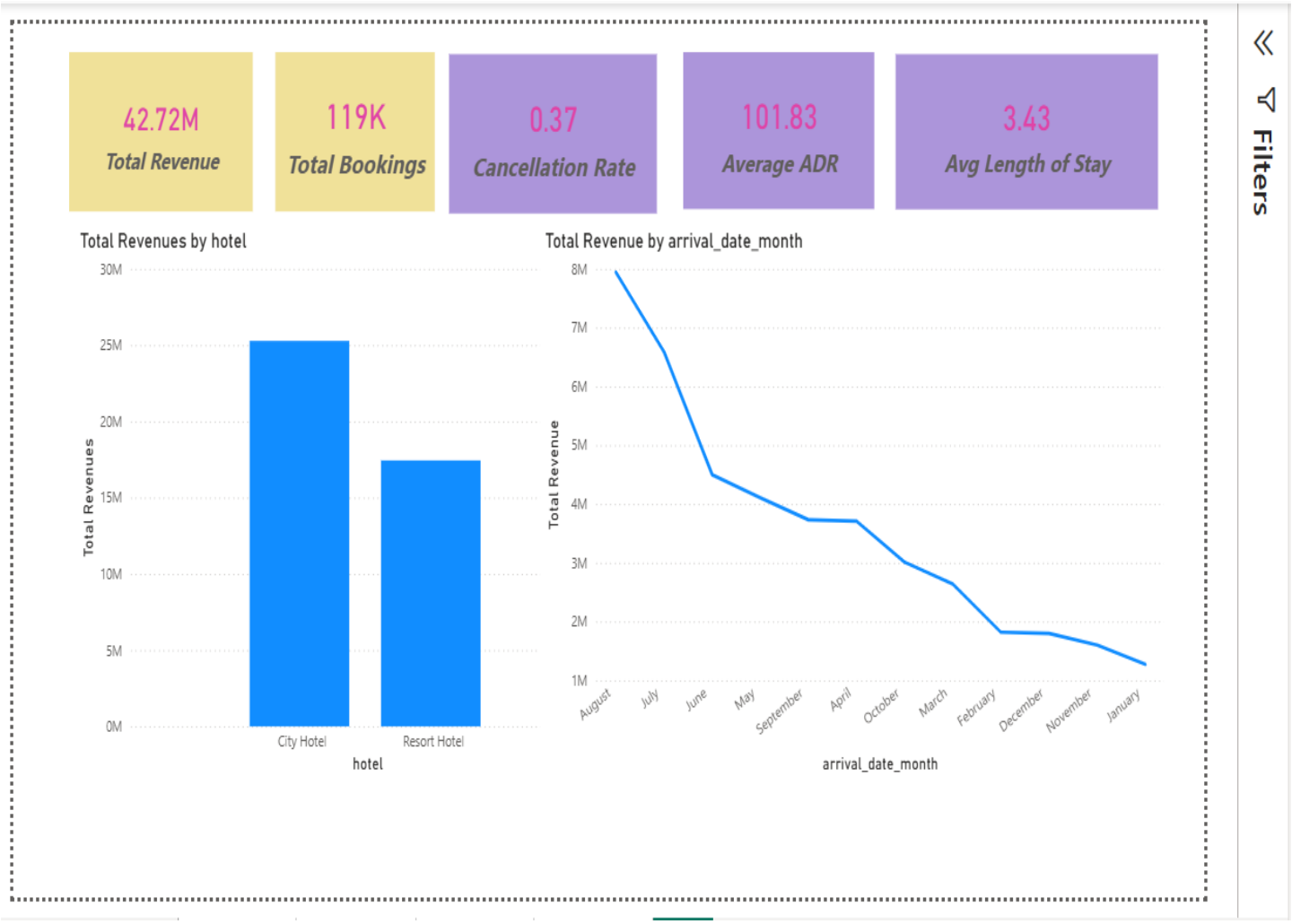
Condition	Action	Reason
adults = 0 AND children = 0	Remove row	Booking with no guests is invalid
adr < 0	Remove row	Negative room rate is invalid
Total Nights = 0	Remove row	Zero-night stay is not a real booking
adr = 0	Investigate	Could be complimentary stay — review before removing
lead_time > 700	Investigate	Very unusual — verify if real booking

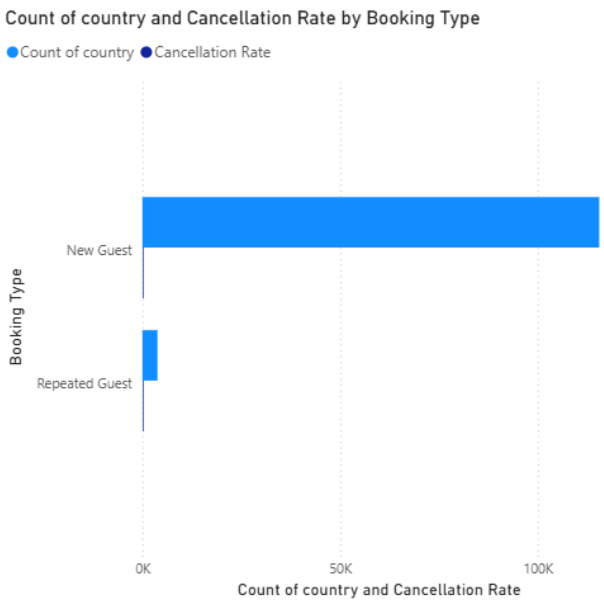
Final Cleaning Checklist

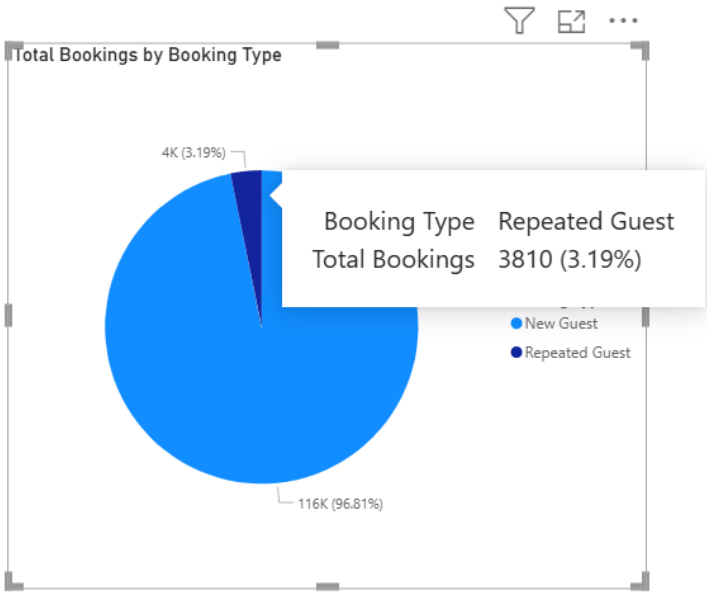
#	Cleaning Step	Status
1	Replace NULL values in agent, company, children	**Done**
2	Fix all column data types (especially Date and Decimal)	**Done**
3	Remove fully duplicate rows (all columns identical)	**Done**
4	Create calculated columns: Total Nights, Total Guests, Cancellation Status, Booking Ty A	**Done**
5	Create DAX measures: Total Revenue, Total Bookings, Cancellation Rate, Average A	R, Avg Length of tay**Done**
6	Trim & Clean all text columns	**Done**
7	Remove invalid rows (no guests, negative ADR, zero nights)	**Done**

Dashboard Structure After Cleaning

Page	Page Title	Key Visuals
1	**Executive Overview**	KPI Cards, Revenue by Month (Line), Revenue by Hotel (Bar), Market Segment (Column), Bo
2	**Cancellation Analysis**	Cancellation Rate by Country (Map/Bar), by Lead Time (Column), Deposit Type vs Cancellatio
3	**Customer Insights**	Repeated vs New Guests (Bar/Donut), Special Requests vs Revenue (Scatter), Parking Spac




Count of country and Cancellation Rate by Booking Type
Count of country
Cancellation Rate
Booking Type
New Guest
Repeated Guest
0K
50K
100K
Count of country and Cancellation Rate


Total Bookings by Booking Type
4K (3.19%)
116K (96.81%)
Booking Type Repeated Guest
Total Bookings 3810 (3.19%)
New Guest
Repeated Guest